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Lexical Reconstruction from Instrumented LLM Token-Length Traces: Reference Validation and Defense Evaluation

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Abstract

Streaming large language model (LLM) services can expose response-length metadata even when payloads are encrypted. We evaluate a fixed T5 implementation with its recorded candidate-selection rule through 6600 reconstructions from validated application-layer length traces. For 300 aligned prompts per model, five archived-reference cohorts yield mean term-frequency cosine scores (ϕ) of 0.482–0.535; two additional archived cohorts have malformed word boundaries. Newly generated TinyLlama and Phi-3.5-mini responses with verified detokenization yield means of 0.435 and 0.504. Legacy rendering of their saved token IDs reproduces all 600 corresponding archived references and length vectors. Changing both the trace and scoring reference increases mean ϕ by 0.356 and 0.383 under matched reconstruction settings. In two-model pilots, three of six partial-ablation contrasts remain significant after Holm correction; neither cross-prompt contrast is conclusive. Five framing transformations reduce frozen-reconstructor mean scores by 36.7–88.7%. A five-seed study with a local empirical maximum-a-posteriori (MAP) length inverse recovers much of the loss under bucketing and batching. Reconstruction scopes and retained encoder coverage differ across these experiments. The findings quantify lexical behavior of an instrumented pipeline and demonstrate the importance of validating both observations and reference rendering. They do not measure semantic disclosure or demonstrate passive extraction from encrypted transport.

Keywords: large language models; information leakage; side-channel analysis; server-sent events; traffic analysis; token-length traces; text reconstruction; privacy

1. Introduction

Interactive large language model (LLM) services stream partial responses to reduce latency. One implementation uses Server-Sent Events (SSE), with generated text carried in application events. Encryption hides the event contents but not necessarily the shape of the communication. If application-event boundaries or correlated record sizes are observable, a sequence of lengths can reveal properties of the underlying tokens.

This paper separates reconstruction from a known length trace from extraction of that trace from encrypted traffic. The laboratory observable is the ordered UTF-8 byte length of each token fragment emitted by an instrumented service, before transport processing. A validated log oracle supplies these lengths and preserves newline bytes and empty fragments. This abstraction removes segmentation by HTTP and Transport Layer Security (TLS), buffering, retransmission, and multiplexing from the experiment. The historical

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line-collector discrepancy motivates the validation in Section 3.4; all quantitative reconstruction results reported below use recorded, validated input vectors. No experiment here establishes passive extraction of event boundaries from encrypted traffic.

Building on Weiss *et al.*, who reconstructed streamed responses from token-length patterns using a T5-based inversion pipeline [1], we examine one fixed reconstructor across open-weight victim models. Partial trace ablations assess its sensitivity to ordered lengths, and framing transformations test its response without retraining. The seven-model cohort retains archived emitted texts; TinyLlama and Phi-3.5-mini are diagnostic within that cohort because of malformed word boundaries. A separate two-model experiment uses newly generated responses with verified full-sequence detokenization.

The study addresses three research questions:

RQ1: Do ordered logged lengths improve lexical scores over controls that shuffle lengths, replace them with a constant, or use another prompt's trace?

RQ2: How do scores from validated length traces differ across seven archived victim-model cohorts, and how do separate experiments with verified reference decoding inform the two diagnostic series?

RQ3: How do bucketing, fixed-width units, batching, and randomized padding affect the evaluated pipeline, protected-payload size, and observable-unit count, and how does a prompt-disjoint maximum-a-posteriori (MAP) inverse change its scores?

The contribution is an evaluation with explicit input provenance: 2100 reconstructions from validated archived lengths, 600 additional reconstructions of newly generated and correctly assembled responses, two-model pilots of partial trace ablations and five framing configurations, and a five-seed MAP-inverse evaluation. The pilots and MAP study use different reconstruction scopes, so their baselines are not interchangeable. Full input vectors, decoder inputs, random seeds, generated texts, and independent metric checks are retained in the accompanying artifact package. The source project is hosted at https://github.com/jackaugustyn/LLeakM. Appendix A identifies the completed experiment artifacts and distinguishes them from historical manifests.

2. Related Work

Side-channel attacks infer sensitive information from indirect observables such as timing and message size. Classical timing attacks showed that implementation-level leakage can defeat cryptographic protection without recovering plaintext directly [2]. Encrypted network applications likewise leak through packet sizes and timing. Keystroke inference, web-application traffic analysis, and website fingerprinting demonstrate that encryption does not hide communication shape [3–6].

Traffic-analysis defenses transform that shape through padding, morphing, batching, or cover traffic. Traffic morphing attempts to make one traffic class resemble another, while subsequent studies show that efficient countermeasures may retain exploitable structure or impose substantial overhead [7,8]. For this reason, defense effectiveness should be evaluated by reporting both the leakage that remains and the resulting bandwidth and latency overhead, rather than by treating padding as either fully protective or ineffective.

For generative AI, Weiss *et al.* introduced T5-based sequence-to-text inversion and, in their first-segment evaluation, reported 27% of reconstructions above an embedding-cosine threshold of 0.9 and approximately 53% above 0.5 [1]. Their cosine uses MiniLM sentence embeddings; its thresholds and first-segment scope are not directly comparable to our term-frequency cosine and full-response LR@0.5. Li *et al.* subsequently studied API outputs that merge several tokens into one observable unit and showed that aggregation does not remove the side channel [9]. Whisper Leak instead infers prompt topics from packet-

size and timing patterns in encrypted LLM traffic [10]. Those studies address stronger deployment-facing threat models. The present work narrows the observation layer to ask a different question: how a fixed reconstructor transfers across victim models once a token-granular length trace is available, and how simple transformations affect that same reconstructor.

The reconstruction pipeline uses T5-style text-to-text transformers [11]. Evaluation combines a reference-normalized Levenshtein distance [12], ROUGE precision [13], and one fixed term-frequency cosine score ϕ. These are lexical measures. They should not be read as semantic correctness or disclosure of a usable answer.

3. Materials and Methods

3.1. Threat Model and Objective

We evaluate a length-only reconstructor for an instrumented LLM service. The observation is the ordered byte length of each emitted token fragment; the attacker receives neither the generating prompt, token identities, nor victim-model internals. The evaluator uses instrumented logs to verify these lengths and to obtain reference text, but supplies only the lengths to the reconstructor. This application-layer oracle does not recover event boundaries from TLS, HTTP/2, or QUIC traffic. The archived-reference cohort and the newly generated two-model cohort are kept separate because verified byte lengths do not themselves establish ordinary detokenization of historical outputs.

The objective is to characterize lexical reconstruction in this controlled apparatus and the response of a fixed reconstructor to length transformations. The experiments capture per-token traces, reconstruct text from length sequences, and compare the reconstructions with the stored emitted responses. Prompts are used to generate those responses and are not supplied to the reconstructor. Figure 1 shows the intended information flow.

3.2. Information-Theoretic Interpretation

Let R denote the stored emitted response and let $X = (L_1, \ldots, L_T)$ denote its ordered token-payload length trace in the idealized laboratory model. When a defense is applied, X is transformed into the observable trace $Z = g(X, N)$, where N is independent transformation randomness. The attacker produces $\widehat{R} = A(Z, U)$ using independent decoder randomness U. Conditional on a fixed fitted attacker, these relationships form the following Markov chain:

$$R \longrightarrow X \longrightarrow Z \longrightarrow \widehat{R}. \tag{1}$$

According to the data-processing inequality,

$$I(R; \widehat{R}) \leq I(R; Z) \leq I(R; X). \tag{2}$$

Neither processing stage increases mutual information about R under these assumptions. The empirical controls are partial trace ablations: they preserve response-dependent features such as trace length, the length histogram, or mean length. A score advantage over an ablation indicates sensitivity to the altered features, but does not isolate all trace information from the decoder's language prior. The lexical score ϕ does not estimate mutual information, and its reduction for a fixed attacker does not measure defended-channel capacity.

For a single-segment reconstruction, let $E(Z)$ denote the deterministic encoder input after inversion (if used), segmentation, serialization and truncation. Then the chain refines to $R \to X \to Z \to E(Z) \to \widehat{R}$, and $I(R; E(Z)) \leq I(R; Z)$. The complete trace Z may retain variation that never reaches the decoder. This distinction matters for `pad_32`: the audited single-call encoder inputs are constant within each evaluated condition (Section 4.6). Multi-

segment decoding additionally uses the previously selected text as context, as specified below.

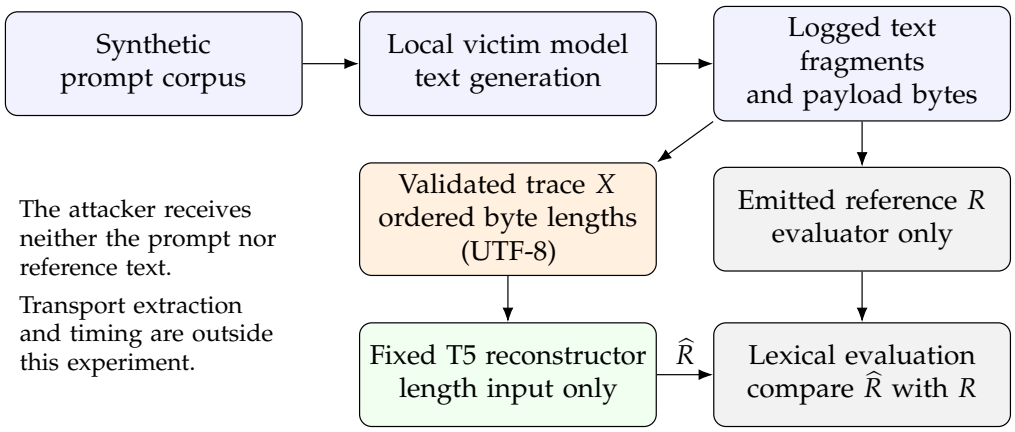


Figure 1. Information flow in the instrumented experiment. Logged fragments supply two separate evaluator products: a validated length trace and the emitted reference text. Only the lengths enter the reconstructor; the reference is used for scoring. The diagram makes no assumption that an instrumented write is a valid SSE event or an observable encrypted-transport record.

3.3. Streaming Harness and Trace Collection

The instrumented harness associates each generated token position with a text fragment and records its UTF-8 length, serialized-write length, and relative timestamp. The historical implementation decoded each token independently; the new-generation implementation uses contextual prefix decoding and verifies that the fragments concatenate exactly to the full decoded token sequence. The new harness completes token generation and contextual decoding before emitting the fragments; it validates payload lengths rather than the timing of incremental generation. Each write has the SSE-like serialization `data: <fragment>\n\n`, and termination uses `data: [DONE]\n\n`. Raw newlines inside fragment text are not escaped, so a write is not necessarily a valid single SSE data event. The present experiments use validated lengths from the instrumented logs, not a line-based parser.

For the logged serialization, the payload length satisfies

$$L_t^{\log} = L_t^{\text{write}} - 8, \tag{3}$$

where the eight bytes are the prefix `data: ` and suffix `\n\n`. This identity concerns the complete emitted byte string, including newline bytes inside the fragment. The terminal `[DONE]` write is excluded. Validation checks per-fragment byte counts, their order, concatenated reference text, total payload bytes, and sample/log identity. For newly collected streams the collector also checks the total received byte count. Exact vectors, source hashes, and reference text are saved with every reconstruction input. The primary, control, defense, and MAP experiments all use this explicit log-length representation.

3.4. Measurement and Reference-Text Audit

The archived collector used `requests.iter_lines()`, discarded empty lines, and appended two newline bytes to each remaining line. Its inferred vector X^{col} can differ from the logged vector $X^{\log}$: a fragment `:\n\n` contains three bytes, but this parser infers one. Replaying that implementation on the retained logs changes 2073 of 2100 vectors and the heuristic-segment count in 764 samples. These are preprocessing comparisons; the exact historical T5 input vectors and their binding to the old collector version were not retained.

The primary results below instead come from complete T5 inference on the 2100 validated logged vectors. This establishes the inputs used for the reported reconstructions without recovering undocumented historical inputs. Differences from older scores cannot be attributed solely to collection: the execution environment and reconstruction random-state handling also differ. The reconstructor is seeded after checkpoint loading so that cache initialization does not change a sample's random stream.

Reference text requires a separate check. The historical harness's independent token decoding produced no ASCII spaces in 299/300 TinyLlama responses and 162/300 Phi-3.5-mini responses. Those 600 archived references remain diagnostic; validating their emitted byte lengths does not restore their word boundaries. Historical token IDs are unavailable for all seven models, so complete-sequence detokenization of the other archived series is also not certified.

To evaluate the affected models with ordinary reference text, we generated 300 additional responses per model from the same prompts using contextual detokenization. All 600 passed exact agreement between concatenated fragments and full-sequence decoding, with token IDs, tokenizer and checkpoint provenance retained; all contain ASCII spaces. These new cohorts are reported separately. Current checkpoint revisions, precision and placement are pinned; their identity with the historical execution is not established.

Holding the newly recorded token IDs fixed, replay of the historical independent-token decoding reproduces every archived fragment, complete response and byte-length vector for all 300 TinyLlama and all 300 Phi-3.5-mini samples. Switching to complete-sequence decoding changes only ASCII word-boundary spaces in these records. The two renderings are reconstructed with the same T5 checkpoints, precision, device, settings and per-sample seeds. This matched comparison measures the joint effect of rendering on the length input and scoring reference. It does not separate those two contributions, recover unknown historical IDs or certify old checkpoint versions.

3.5. Reconstruction Pipeline

The inversion pipeline follows the sequence-to-text procedure introduced by Weiss *et al.* [1]. It first splits a length vector at observations equal to one and merges short chunks using a ten-unit minimum, with an additional list-boundary heuristic. These are heuristic segments rather than verified sentences. The first segment is decoded using `royweiss1/T5_FirstSentences`; later segments use `royweiss1/T5_MiddleSentences`, conditioned on the immediately preceding selected segment. T5 generation uses stochastic sampling with top-$k = 50$, temperature 1 and a two-token no-repeat n-gram constraint. The executed selector sums normalized transition scores over all returned generation steps, including PAD positions after a candidate's first end-of-sequence (EOS) token; the chosen text supplies the next segment's context.

These scores are obtained after the sampling constraints, including top-k and the no-repeat rule; their sum is not an unconstrained model likelihood. The implementation does not mask post-EOS steps, which can assign nonfinite scores to a completed candidate and alter selection. A bounded replay reproduced all 24 selected first-segment predictions, but ranking only through the first EOS changed the selected text in 18 cases (Appendix C). All three-candidate results in this paper retain the original PAD-inclusive selector. They describe this specific implementation and do not estimate the performance of an EOS-masked selector. The one-candidate MAP experiment bypasses candidate ranking.

The pinned model cards identify these checkpoints as the T5-Large fine-tunes used by Weiss *et al.*, trained on the first answers from UltraChat's "Questions About the World" subset. The source procedure trains separate models for first segments and subsequent segments, with preceding-segment text providing context for the latter [1]. We verify the

downloaded checkpoint revisions but do not independently reproduce training or audit its underlying records. The original paper defines token lengths in characters; our traces measure UTF-8 bytes of emitted fragments. These units agree for ASCII text but can differ for non-ASCII text, adding a representation shift alongside differences in tokenizer and response style.

For defense experiments, the order is length transformation, optional MAP inversion, heuristic segmentation, serialization of each segment as underscore-prefixed integers, T5 tokenization, input truncation, and decoding. The encoder input is limited to 80 T5 tokens, including the instruction prefix and any preceding context; generation also uses `max_length`=80. These limits are distinct from the victim's 96- or 512-token generation budget. One length value need not occupy one T5 token: for example, _32 occupies two tokens in the audited tokenizer. Changing the length values can therefore change both segmentation and the amount of trace retained by the encoder. Section 3.11 reports the audited coverage of the first segment in the MAP experiment.

The reconstruction design is conditional rather than purely tokenwise. Instead of recovering each token independently, it maps length patterns into coherent text spans and uses context to improve linguistic plausibility (Figure 2). Our quantitative claims nevertheless concern lexical recovery; semantic correctness was not established by human annotation.

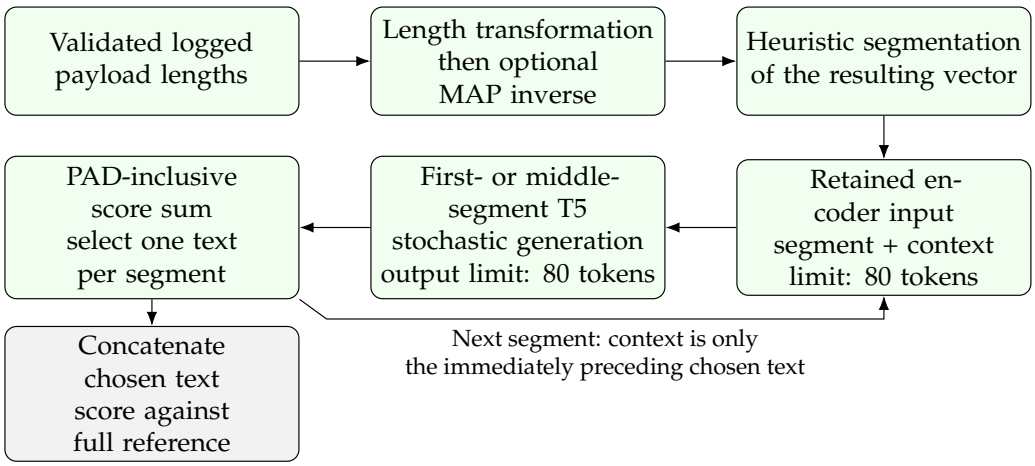


Figure 2. Executed reconstruction pipeline. Unmodified conditions pass lengths directly to segmentation. The original PAD-inclusive selector precedes each context update (Section 3.5); the first segment has no reconstructed context. Main and new-reference runs decode all segments with three samples per segment; 96-token pilots allow up to three segments with three samples each; the five-seed MAP study uses one sample for the first segment per seed. The 80-token encoder limit includes the instruction and context. All settings score their output against the full stored response.

3.6. Prompt Corpus and Experimental Protocol

Table 1 summarizes the cohorts, reconstruction scopes and seed rules.

The synthetic prompt corpus comprised 15 topical files with 20 prompts each, yielding 300 distinct prompts. It covered legal, health-related, financial, employment, and security matters. The generator combines 60 topic–scenario pairs with five wording templates; the 300 prompts therefore do not represent 300 independent scenarios. Prompt scheduling followed a round-robin strategy to reduce topical clustering during data collection.

Victim-model decoding used greedy selection, with temperature 0.0 and `top_p`=1.0. The primary cohort contains $N = 300$ archived responses per model generated with a 512-token budget. Each reconstruction requests all heuristic segments, with `max_sentences`=0. It samples three candidates per segment and retains up to three first-segment candidates.

Table 1. Protocol map. *N* counts reconstructions, including repeated conditions and decoder seeds; it is not a count of independent victim responses. All experiments score against the full reference and impose 80-token T5 encoder and decoder limits. *All* requests every heuristic segment; the pilot cap is three segments. The controls, defense pilot and MAP study use Qwen2.5-1.5B and Llama-3.2-3B. Samples are candidates per segment, not independent victim responses. The last column refers to the seed rules in Appendix B.

Experiment / result tables	References and victims	*N*	Victim limit	Segments / samples	Decoder seeds
Archived main; 6–9	Archived emitted text; seven models, 300 per model; two diagnostic series	2100	512	All / 3	Recorded
Verified rendering; 10	New full-sequence decoding; 300 per model, TinyLlama and Phi-3.5-mini	600	512	All / 3	Recorded
Trace controls; 5	Archived text; two models, 30×4 per model	240	96	≤ 3 / 3	Pilot
Defense pilot; 11	Archived text; two models, 30×6 per model	360	96	≤ 3 / 3	Pilot
MAP study; 12	Archived text; two models, $30 \times 5 \times 11$ per model	3300	512	First / 1	Common, five

Six historical victim models used a one-token KV-cache loop, while Gemma-2-2B-it used greedy Hugging Face `model.generate()`. Responses reaching EOS before the limit are marked `response_complete`; length-truncated responses remain in the primary analysis. EOS-complete subsets are descriptive subsets, not interchangeable samples.

The evaluated models are listed in Table 2. Six are instruction- or chat-tuned; Phi-1.5 is a base-model stress test and does not identify a general base-versus-instruction effect [14–20]. A common token budget does not imply a common text length across tokenizers. Table 3 describes the 2100 archived emitted responses, which are unchanged by reconstruction. Their truncation rates range from 34.3% to 100%; archived Phi-3.5-mini has no EOS-complete response.

The additional TinyLlama and Phi-3.5-mini generations use the same 300 prompts, the legacy prompt format, greedy decoding and a 512-token limit. TinyLlama revision `fe8a4ea1ffedaf415f4da2f062534de366a451e6` is loaded in float16 on CUDA; Phi-3.5-mini revision `2fe192450127e6a83f7441aef6e3ca586c338b77` uses float16 with automatic CPU offload to fit the available memory. Reconstruction for every cohort uses float32 T5 checkpoints on an NVIDIA GeForce RTX 3070 Ti Laptop GPU with 8 GB VRAM, PyTorch 2.5.1 and Transformers 4.46.3. Exact model files, revisions and environment details are bound to each protocol in the artifacts.

Table 2. Victim checkpoints and tokenizer families in the archived-reference cohort. TinyLlama and Phi-3.5-mini are diagnostic in that cohort and also evaluated separately with newly generated, validated references. Parameter counts follow the cited model cards and technical reports; these metadata do not certify historical checkpoint revisions.

Model	Checkpoint	Parameters	Tokenizer
Qwen2.5-1.5B-Instruct	`Qwen/Qwen2.5-1.5B-Instruct`	1.54B	Qwen2
Qwen2.5-3B-Instruct	`Qwen/Qwen2.5-3B-Instruct`	3.09B	Qwen2
Llama-3.2-3B-Instruct	`meta-llama/Llama-3.2-3B-Instruct`	3.2B	Llama 3
Gemma-2-2B-it	`google/gemma-2-2b-it`	2.6B	Gemma
Phi-1.5	`microsoft/phi-1_5`	1.3B	CodeGen
TinyLlama-1.1B-Chat-v1.0	`TinyLlama/TinyLlama-1.1B-Chat-v1.0`	1.1B	Llama 2
Phi-3.5-mini-instruct	`microsoft/Phi-3.5-mini-instruct`	3.8B	Llama/Phi-3

Table 3. Archived response-length covariates ($N = 300$ per model), including the two diagnostic series. These describe the source responses used by the completed reconstructions. Character and byte counts refer to emitted token text, excluding framing bytes; mean B/token is averaged over responses. Truncation is the percentage of samples with `token_count` ≥ 512.

Model	Mean tokens	Mean chars	Mean B/token	Trunc. (%)
Qwen2.5-1.5B-Instruct	418.3	2069	4.91	44.3
Qwen2.5-3B-Instruct	497.6	2417	4.87	83.3
Llama-3.2-3B-Instruct	431.0	2081	4.84	34.3
Gemma-2-2B-it	490.6	2257	4.60	74.3
Phi-1.5	395.4	2035	5.11	52.0
TinyLlama-1.1B-Chat-v1.0	428.3	1478	3.45	41.0
Phi-3.5-mini-instruct	512.0	1755	3.43	100.0

3.7. Evaluation Metrics

Reconstruction quality was computed against the complete reference response R using four lexical criteria. The normalized Levenshtein value is `ed_norm` $= d(R, \widehat{R})/|R|$, using reference character length as the denominator; it can exceed 1 when the prediction is much longer than the reference [12]. ROUGE-1 precision and ROUGE-L precision measure unigram and longest-common-subsequence overlap [13]. The primary metric ϕ is cosine similarity between unweighted, case-normalized term-frequency vectors [21], with terms defined by `[A-Za-z0-9']+`. All reported metrics were recomputed from complete saved text, using independent edit-distance and subsequence implementations checked against dynamic programming. The MAP experiment scores the first heuristic segment's prediction against the entire response, not against an annotated first sentence.

The study also reports lexical recovery rate LR@0.5, the proportion of samples with $\phi > 0.5$. This threshold is a descriptive operating point, not a calibrated success criterion. Neither ϕ nor LR@0.5 establishes factual, semantic, or operationally useful reconstruction.

3.8. Trace-Information Controls

We compared four inputs on the same topic-balanced set of 30 prompts for Qwen2.5-1.5B-Instruct and Llama-3.2-3B-Instruct using direct logged lengths from the 96-token pilot: (i) the original trace; (ii) its lengths randomly permuted within the response; (iii) a constant sequence equal to its rounded mean length; and (iv) another prompt's trace cycled to the target trace's number of positions. These are partial trace ablations. Shuffling preserves the response's length histogram and number of positions, the constant condition preserves its mean-derived value and number of positions, and the cross-prompt condition preserves the target trace's number of positions while replacing the length pattern. Thus each condition retains some response-dependent information. The comparisons probe sensitivity to order, variation, and alignment; their residual scores cannot be assigned wholly to the language prior. The control experiment reconstructs up to three heuristic segments against the full stored response and saves the actual segment counts and encoder inputs. These comparisons are not estimates of mutual information.

3.9. Statistical Analysis

Because the same 300 prompts were issued to every model, comparisons of the retained scores used prompt-aligned observations. For the archived main and verified-reference cohorts, percentile-bootstrap 95% confidence intervals [22] use $B = 10{,}000$ resamples from a generator initialized with seed `20260706`; the pinned aggregator consumes this generator in its recorded model, metric and contrast order. The pilot and MAP intervals use separately derived cell and contrast seeds, specified in Appendix B. Confidence intervals for mean

pairwise differences were obtained by bootstrapping within-prompt differences. Wilson 95% intervals were used for LR@0.5. These nominal intervals describe variation across the corpus's prompt variants under prompt-level resampling. They neither account for dependence among the five variants of a scenario nor establish generalization to unseen scenarios, and they do not include repeated victim generations or unresolved measurement errors. A separate exploratory sensitivity analysis averages the five variants within each of the 60 scenarios, resamples those scenario means 10,000 times from the same aggregator's random-number stream, and applies paired signed-rank tests to the 60 aligned means with Holm correction over 21 pairs. It changes the resampling and testing unit while preserving the equal-weight corpus mean; it does not establish generalization to unseen scenarios or repair measurement error. Decoder-seed uncertainty is reported separately for a two-model subset (Section 3.11).

Pairwise differences in per-prompt ϕ were assessed using two-sided paired Wilcoxon signed-rank tests [23]. Absolute nonzero differences receive average ranks for ties; zero differences are discarded. The normal approximation uses a tie-adjusted variance and a continuity correction of 0.5, with normal tails evaluated directly to avoid numerical cancellation. Holm–Bonferroni correction is applied separately to 21 model comparisons, six control–true-trace comparisons and ten defense–baseline comparisons [24]; rank-biserial correlation summarizes effect size. Significance denotes adjusted $p < 0.05$. A bootstrap interval for a mean difference and a multiplicity-adjusted signed-rank test address different summaries and are not treated as interchangeable.

3.10. Defense Evaluation Protocol

Five configurations were evaluated on the same 30 topic-balanced prompts for each of Qwen2.5-1.5B-Instruct and Llama-3.2-3B-Instruct, using direct logged lengths from earlier 96-token traces. These use different source responses and a lower victim-token budget than the primary cohort, with the same validated log-length representation. Let ℓ_t denote the logged token-payload byte length. The pilot reconstructs up to three heuristic segments, with three samples per segment, and scores their concatenation against the full stored response.

- `bucket_8`: replace ℓ_t by $8\lceil \ell_t/8 \rceil$, independently and online for every event.
- `pad_32`: replace each token-length observation by a constant 32-byte protected unit. The largest token in the evaluated subset occupied 16 UTF-8 bytes, so every token fit within the unit.
- `batch_2` and `batch_4`: sum the lengths of consecutive groups of two or four tokens into one attacker-visible value. The final incomplete group, if present, is retained.
- `rand_pad_8`: add an independently sampled integer $U_t \sim \text{Uniform}\{0, \ldots, 8\}$ using the pilot seed recipe described below.

The transformed sequences were passed through segmentation, T5 tokenization, input truncation, and decoding. The reconstructor was frozen and was not retrained or calibrated on defended traces. We report recovery-score reduction as $100(1 - \bar{\phi}_d/\bar{\phi}_0)$. Payload-size overhead compares the sum of transformed units with the sum of original token bytes. This deliberately narrow accounting excludes SSE syntax, transport headers, packetization, buffering delay, and user-perceived latency. Batching has zero overhead only under this payload-sum definition; its reduction in observable unit count is reported separately.

The control and defense experiments use explicit BLAKE2s-derived, condition-specific decoder seeds and record every seed and transformed vector. Shuffling uses seed $20260706 + 17i$ for prompt index i. Cross-prompt inputs cycle the trace of donor index $(i \bmod 30) + 1$ to the target's trace length. Random padding has a separately recorded transformation seed, and its payload overhead is calculated from the actual saved padding

realization. These are newly specified pilot random draws, not recovered historical seeds. The control and defense baseline reconstructions use distinct decoder seeds, so even their identical undefended length traces can yield different mean scores. Historical `hash()`-dependent seeds and inconsistent padding-cost summaries are excluded from the current results.

3.11. Repeated Decoder Seeds and Adaptive Attacker

The primary seven-model table uses one assembled reconstruction per sample, selecting candidates from three stochastic samples per segment. Decoder-seed uncertainty and a defense-aware attacker were evaluated using direct logged lengths from the 512-token runs of Qwen2.5-1.5B-Instruct and Llama-3.2-3B-Instruct. Two prompts were taken from each of the 15 topics ($n = 30$ test prompts per model); the remaining 270 traces formed a calibration set with no identical test prompt. All 30 test scenarios nevertheless occur in 120 calibration rows under the other four wording templates. The test uses the first wording template throughout. Thus the split is exact-prompt-disjoint, but not scenario-disjoint. An EOS-complete-only split would have left 137 (Qwen) and 167 (Llama) calibration traces, below the configured minimum of 200, and was therefore not used. Of the 30 test prompts, 24 (Qwen) and 21 (Llama) were EOS-complete.

Reconstruction was repeated for five decoder seed indices, `20260706–20260710`. For a given prompt the same decoder seed was used in every defense and attacker condition (common random numbers). The seed is derived by BLAKE2s over the tuple consisting of `decoder`, the seed index, and the prompt index and does not depend on `PYTHONHASHSEED`. Mean ϕ is reported with a crossed bootstrap 95% confidence interval that resamples the prompt axis and the seed axis independently ($B = 10{,}000$). Between-seed variation is the sample standard deviation of the five seed-specific means; seed repeats are not treated as independent prompts. These reconstructions use one stochastic T5 sample for the first heuristic segment per seed, scored against the full stored response. The primary table instead reconstructs every available segment with three samples per segment; its scores and the MAP scores therefore have different reconstruction scopes.

The adaptive attacker applies a local empirical MAP length inverse before the unchanged T5 decoder. Calibration uses the 270 traces of the same victim model with no exact test prompt. The inverse selects a modal length conditional on an observed value, or a modal length tuple conditional on a batch sum and group size; unseen observations and the final batch use explicit fallback and size-selection heuristics (Appendix B). This is not a global MAP estimate of the response's entire length sequence. This is a calibrated length inverse, not end-to-end T5 retraining. Baseline traces are scored with the frozen reconstructor only. Each of the five transformations is scored with both the frozen reconstructor and the MAP inverse. The design comprises 22 complete cells, each with 30 prompts and five decoder seeds. Source vectors, inverse inputs, and decoder seeds are retained; the complete experiment contains 3300 reconstructions.

The recorded encoder inputs quantify first-segment coverage (Table 4). Baseline first segments have median lengths of 14.5 observed units for Qwen and 16 for Llama, all retained. For `bucket_8`, median segments contain 451.5 and 441.5 units, but only the first 64 units fit after the instruction prefix. For `pad_32`, 32 units fit because each length value occupies two T5 tokens. With MAP inversion, all 30 prompt-indexed inputs per model remain truncated for bucketing and fixed padding; counts are one per model for `batch_2`, zero for `batch_4`, and eight (Qwen) and two (Llama) for `rand_pad_8`. Five decoder seeds reuse each condition's input. Actual saved token IDs were checked against the pinned tokenizer. The comparisons therefore combine changes in length values, segmentation, encoded representation, and retained prefix; observation coverage is not held constant.

Table 4. Recorded first-segment encoder inputs exceeding 80 T5 tokens in the frozen-attacker conditions of the five-seed experiment (30 test prompts per model). Each condition is repeated for five decoder seeds; identical encoder inputs can occur for different prompts. The limit includes the instruction prefix. Counts describe encoder coverage, not reconstruction quality.

Condition	Qwen2.5-1.5B	Llama-3.2-3B
baseline	0/30	0/30
bucket_8	30/30	30/30
pad_32	30/30	30/30
batch_2	30/30	30/30
batch_4	30/30	29/30
rand_pad_8	18/30	9/30

4. Results

The completed experiments comprise 2100 reconstructions from validated archived length traces, 600 reconstructions of newly generated TinyLlama and Phi-3.5-mini responses, 240 trace-control reconstructions, 360 defense-pilot reconstructions and 3300 MAP reconstructions. Every planned sample is present and independently checked. The primary and new-reference cohorts reconstruct all available heuristic segments; the 96-token pilots allow up to three, and the 512-token MAP experiment reconstructs one. These differences preclude interpreting cross-experiment score differences as victim-budget effects alone.

4.1. Sensitivity to Partial Trace Ablations

Table 5 shows that the effect of removing trace features depends on both model and control. For Qwen, the true-trace mean is 0.2904 versus 0.2084 after shuffling and 0.1169 for a constant trace; both contrasts are significant after Holm correction ($p = 0.0053$ and $p < 0.0001$). For Llama, the true mean is 0.2221 versus 0.1272 for a constant trace ($p = 0.0010$). Its shuffled mean is lower (0.1798), but the adjusted signed-rank test is inconclusive ($p = 0.2634$). Cross-prompt means are 0.2522 for Qwen and 0.2250 for Llama; neither contrast is significant ($p = 0.3625$ and $p = 0.8531$). Thus these pilots do not establish a general advantage over unrelated prompt traces. Each control retains some response-dependent information, and transformations can also alter segmentation and encoder coverage.

Table 5. Partial trace ablations using logged token-byte lengths from the 96-token pilot ($n = 30$ prompts per model and condition). Each run decodes up to three heuristic segments with three samples per segment, 80-token T5 input and output limits, and the full stored response as reference. Intervals are percentile-bootstrap 95% confidence intervals for the mean.

Model	Trace	Mean ϕ	95% CI	Difference from true
Qwen2.5-1.5B	true	0.2904	[0.2471, 0.3362]	–
	shuffled	0.2084	[0.1763, 0.2426]	−0.0819
	constant	0.1169	[0.0977, 0.1389]	−0.1735
	cross-prompt	0.2522	[0.2166, 0.2877]	−0.0382
Llama-3.2-3B	true	0.2221	[0.1922, 0.2535]	–
	shuffled	0.1798	[0.1555, 0.2024]	−0.0424
	constant	0.1272	[0.1066, 0.1494]	−0.0949
	cross-prompt	0.2250	[0.1944, 0.2564]	+0.0028

4.2. Reconstruction from Validated Archived Lengths

Table 6 reports the completed reconstruction cohort, and Figure 3a shows its mean scores with scenario-bootstrap intervals. For the five series without the observed malformed-word-boundary pattern, mean ϕ ranges from 0.4820 to 0.5355 and LR@0.5 from 42.33% to 65.33%. Their ROUGE-1 precision is 0.2101–0.2921. These metrics quantify lexical overlap with the archived emitted text; they do not establish semantic fidelity or independently validate its historical full-sequence detokenization. Phi-1.5 remains a base-model stress test.

The archived TinyLlama and Phi-3.5-mini references yield means of 0.0792 and 0.1210, with no $\phi > 0.5$ cases. These are reference-format diagnostics. The new input vectors are verified byte-for-byte for all models, but reference defects remain part of these two scores. Across the 2100 reconstructions, all 62,757 available heuristic segments were processed; 1475 encoder calls exceeded the 80-token input limit. No run stopped early because of an empty middle-segment output.

Table 6. Completed reconstruction runs from validated archived payload lengths ($N = 300$ per model; 512-token victim budget; all available heuristic segments; three samples per segment; 80-token T5 input and output limits). Full archived emitted responses are the references. The final two rows are reference-format diagnostics; input provenance is verified for every row.

Model	Type	ϕ_{mean}	ed_{norm}	R1 Precision	RL Precision	LR@0.5
Qwen2.5-1.5B-Instruct	instruct	0.5355	0.7042	0.2921	0.1760	65.33%
Qwen2.5-3B-Instruct	instruct	0.5251	0.7431	0.2556	0.1302	63.00%
Phi-1.5	base	0.5164	0.7229	0.2566	0.1524	56.67%
Llama-3.2-3B-Instruct	instruct	0.5146	0.7462	0.2464	0.1313	61.00%
Gemma-2-2B-it	instruct	0.4820	0.7748	0.2101	0.1035	42.33%
Reference-format diagnostics						
Phi-3.5-mini-instruct	instruct	0.1210	0.8781	0.0272	0.0202	0.00%
TinyLlama-1.1B-Chat	chat	0.0792	0.8803	0.0282	0.0211	0.00%

Relative to the earlier stored reconstructions, mean ϕ changes by -0.035 to -0.021 for the five non-diagnostic series, while the diagnostic TinyLlama and Phi-3.5-mini means increase. This is a comparison of reconstruction executions on the same archived references, not an isolated estimate of a collector effect. The reported primary metrics use complete new predictions and a single lexical scoring implementation.

4.3. Completeness: All Samples versus EOS-Only

Table 7 compares all 300 archived responses per model with the subset whose victim generation reached EOS. The generation-completeness rates are unchanged because reconstruction uses the same references. EOS restriction does not systematically increase ϕ: Qwen2.5-3B rises from 0.5251 to 0.5551 ($n = 50$), whereas Phi-1.5 falls from 0.5164 to 0.4868 ($n = 144$). Phi-3.5-mini has no EOS-complete subset. EOS indicates completion of victim generation, not correct reconstruction; conditioning on it also changes the evaluated response subset.

4.4. Statistical Uncertainty and Pairwise Tests

Table 8 gives prompt-bootstrap intervals and Wilson intervals for the lexical threshold. Fifteen of 21 model pairs are significant after Holm correction, with the same decisions when the 60 scenario means are used as the paired units. Among Qwen2.5-1.5B, Qwen2.5-3B, Llama-3.2-3B and Phi-1.5, all six pairwise tests are inconclusive after correction. The significant family also contains comparisons with malformed-reference diagnostics, which cannot be interpreted as transfer differences on ordinary model responses.

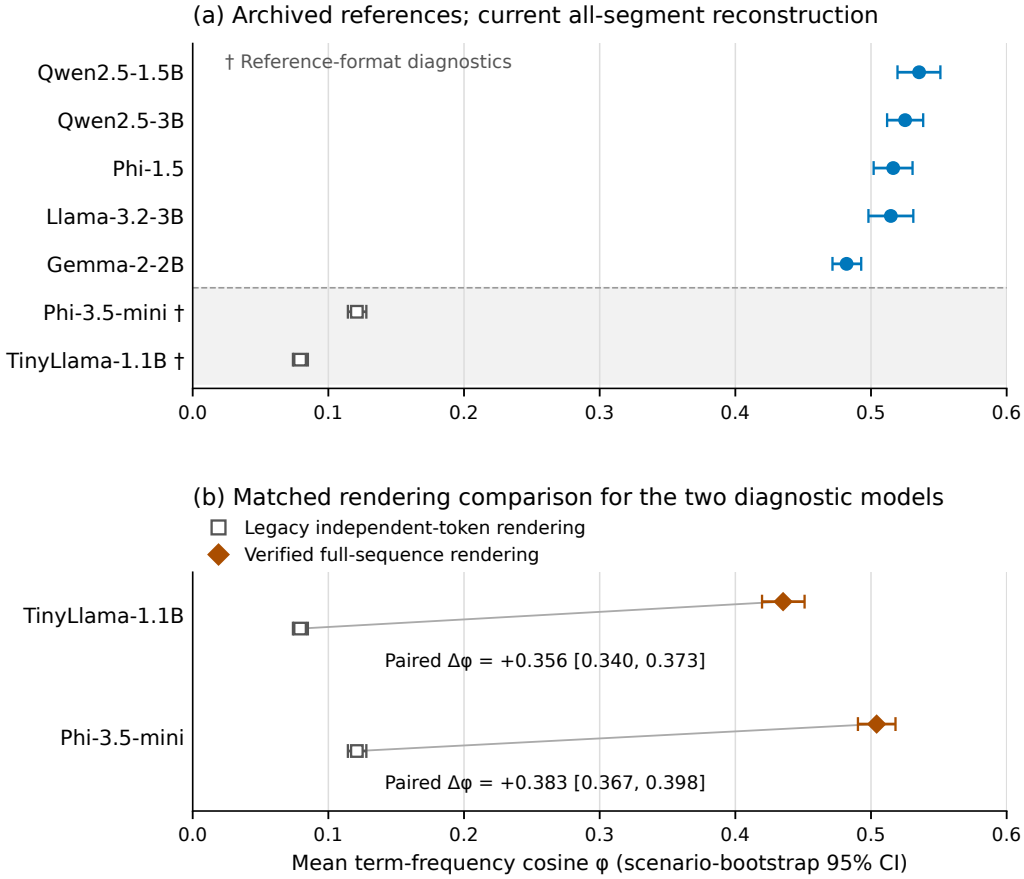


Figure 3. Completed all-segment reconstructions with scenario-bootstrap 95% intervals ($N = 300$ prompts in 60 scenarios per model; 512-token victim limit; three T5 samples per segment; 80-token T5 input and output limits). (a) Archived-reference results; † marks reference-format diagnostics. (b) Separate, matched rendering experiment: legacy decoding of newly recorded token IDs reproduces the two archived diagnostic inputs and references exactly. Full-sequence decoding changes both the trace and scoring reference under identical T5 settings and seeds. Annotated $\Delta\phi$ intervals resample paired scenarios. Marginal intervals match Tables 9 and 10. The two panels do not form a pooled seven-model ranking.

For example, Qwen2.5-1.5B minus Qwen2.5-3B has a mean difference of 0.0103 (prompt-bootstrap 95% CI $[-0.0031, 0.0232]$, rank-biserial $r = 0.1057$, $p_{\text{Holm}} = 0.4500$). Qwen2.5-1.5B minus Llama has a mean difference of 0.0208 (95% CI $[0.0059, 0.0359]$, $r = 0.1408$), but its multiplicity-adjusted signed-rank test is inconclusive ($p_{\text{Holm}} = 0.2076$). These results illustrate why unadjusted mean-difference intervals and adjusted rank tests should be distinguished.

The exploratory scenario analysis retains the same 15 significant pairs (Table 9). All seven scenario-bootstrap intervals are wider than their prompt-bootstrap counterparts, by up to 41.6% for Qwen2.5-3B. For the Qwen2.5-1.5B versus Qwen2.5-3B comparison, the mean difference remains 0.0103, with scenario-level $r = 0.2470$ and $p_{\text{Holm}} = 0.3876$. Grouping the five variants changes the inference unit; agreement of significance decisions does not establish independence or generalization to unseen scenarios.

4.5. Separate Experiments with Verified Reference Decoding

New TinyLlama and Phi-3.5-mini generations passed full-sequence detokenization checks for all 300 prompts per model. Their corresponding mean scores are 0.4352 and 0.5042 (Table 10 and Figure 3b), with LR@0.5 of 30.33% and 52.33%. TinyLlama reached

Table 7. EOS subsets of the archived-reference cohort ($N = 300$ per model; validated length inputs, all segments, 80-token T5 input and output limits). EOS-only metrics select `response_complete` samples; they do not certify correct reconstructions. Both LR columns report percentages. Phi-3.5-mini and TinyLlama remain reference-format diagnostics.

Model	n_{EOS}	EOS (%)	Trunc. (%)	ϕ all	ϕ EOS	LR all	LR EOS
Qwen2.5-1.5B-Instruct	167	55.7	44.3	0.5355	0.5528	65.33	71.26
Qwen2.5-3B-Instruct	50	16.7	83.3	0.5251	0.5551	63.00	74.00
Phi-1.5	144	48.0	52.0	0.5164	0.4868	56.67	40.97
Llama-3.2-3B-Instruct	197	65.7	34.3	0.5146	0.5018	61.00	56.35
Gemma-2-2B-it	77	25.7	74.3	0.4820	0.4743	42.33	44.16
Phi-3.5-mini-instruct	0	0.0	100.0	0.1210	–	0.00	–
TinyLlama-1.1B-Chat	177	59.0	41.0	0.0792	0.0853	0.00	0.00

Table 8. Intervals for reconstructions from validated archived lengths ($N = 300$ per model; all segments, 80-token T5 input and output limits). Mean-ϕ intervals use 10,000 prompt-index bootstrap resamples; LR@0.5 intervals are Wilson 95% intervals. These nominal intervals do not include shared-scenario dependence or reference-format error. The final two rows remain diagnostic.

Model	Mean ϕ	95% CI	LR@0.5	95% CI
Qwen2.5-1.5B-Instruct	0.5355	[0.5235, 0.5477]	65.33%	[59.78, 70.49]%
Qwen2.5-3B-Instruct	0.5251	[0.5156, 0.5344]	63.00%	[57.40, 68.27]%
Phi-1.5	0.5164	[0.5039, 0.5288]	56.67%	[51.01, 62.15]%
Llama-3.2-3B-Instruct	0.5146	[0.5022, 0.5266]	61.00%	[55.37, 66.35]%
Gemma-2-2B-it	0.4820	[0.4735, 0.4904]	42.33%	[36.87, 47.99]%
Phi-3.5-mini-instruct	0.1210	[0.1148, 0.1274]	0.00%	[0.00, 1.26]%
TinyLlama-1.1B-Chat	0.0792	[0.0741, 0.0845]	0.00%	[0.00, 1.26]%

EOS in 177/300 responses; all Phi-3.5-mini responses reached the 512-token limit. All 8835 and 10,748 available reconstruction segments were processed, with 55 and 77 encoder calls truncated, respectively.

Because legacy decoding of the newly saved IDs exactly reproduces the archived references and vectors, the two renderings support a matched computational comparison with identical T5 settings and per-sample seeds. Changing both the length trace and reference to full-sequence rendering increases mean ϕ by 0.3560 for TinyLlama (paired scenario-bootstrap 95% CI $[0.3396, 0.3731]$) and 0.3831 for Phi-3.5-mini ($[0.3670, 0.3982]$). These intervals use 10,000 resamples of the 60 paired scenario means with seed `20261101`. The effect combines changed attacker input and changed scoring text; neither component is isolated. The two cohorts remain separate from the other five archived-reference series, whose original token IDs are unavailable.

4.6. Effects of Framing Transformations on the Frozen Reconstructor

All five framing transformations reduce mean ϕ relative to their defense-pilot baseline (Table 11), and all ten paired comparisons are significant after Holm correction. Reductions range from 36.7% for `pad_32` on Qwen to 88.7% for `bucket_8` on Qwen. Batching reduces scores by 67.2–73.2% and changes the number of observable units from 96 to 48 or 24. Figure 4 relates these changes to payload overhead calculated from the actual saved transformation vectors.

These results describe the complete frozen pipeline, including changes in heuristic segmentation and encoder truncation after each transformation. They do not isolate information removed by a defense. In all 60 `pad_32` pilot records, the retained encoder input is identical and reconstruction makes one decoder call. An extended audit covering these records and all 600 `pad_32` records in the five-seed study finds one unique retained

Table 9. Exploratory scenario-level sensitivity of the validated-length reconstructions: five prompt variants averaged within each of 60 scenarios per model, with 10,000 scenario-bootstrap resamples. These intervals concern the corpus mean; † denotes a reference-format diagnostic.

Model	Mean ϕ	Scenario-bootstrap 95% CI
Qwen2.5-1.5B	0.5355	[0.5195, 0.5511]
Qwen2.5-3B	0.5251	[0.5119, 0.5385]
Phi-1.5	0.5164	[0.5019, 0.5306]
Llama-3.2-3B	0.5146	[0.4981, 0.5312]
Gemma-2-2B	0.4820	[0.4716, 0.4928]
Phi-3.5-mini†	0.1210	[0.1144, 0.1280]
TinyLlama-1.1B†	0.0792	[0.0739, 0.0849]

Table 10. Separate cohorts with newly generated, verified full-sequence references ($N = 300$ per model; 512-token victim limit; all segments, three T5 samples per segment and 80-token T5 input and output limits). Mean-ϕ intervals use 10,000 resamples of 60 scenarios. These rows form a targeted reference-validation experiment and are analyzed separately from the seven archived-reference cohorts. EOS indicates end-of-sequence termination, not reconstruction correctness.

Model	Mean ϕ	Scenario 95% CI	ed_{norm}	R1 Precision	RL Precision	LR@0.5	EOS
TinyLlama-1.1B	0.4352	[0.4197, 0.4510]	0.7576	0.2209	0.1269	30.33%	59.0%
Phi-3.5-mini	0.5042	[0.4904, 0.5181]	0.7733	0.2240	0.1084	52.33%	0.0%

input within each suite, model and attacker group. Although the full 512-token-budget traces differ in their number of positions, that variation is removed before decoding by the 80-token encoder limit. The positive lexical scores therefore cannot demonstrate this decoder's use of response-specific trace-length variation. They reflect overlap between the references and outputs generated from the constant retained input under the recorded seeds; another attacker could still use information in the complete observable trace.

Table 11. Framing transformations on validated lengths from 96-token responses ($n = 30$ paired prompts per model; three samples per segment, up to three segments, 80-token T5 input and output limits, full-response references). Every overhead, including randomized padding, is the realized cost of the saved transformed vectors. Units count entries in the transformed length vector. Payload accounting excludes SSE syntax, transport headers, buffering delay and latency.

Model	Defense	Mean ϕ	95% CI	Score reduction	Payload overhead	Units
Qwen2.5-1.5B	baseline	0.2814	[0.2399, 0.3268]	–	0.0%	96
	bucket_8	0.0318	[0.0224, 0.0421]	88.7%	82.4%	96
	pad_32	0.1781	[0.1515, 0.2054]	36.7%	538.0%	96
	batch_2	0.0754	[0.0613, 0.0911]	73.2%	0.0%	48
	batch_4	0.0844	[0.0626, 0.1071]	70.0%	0.0%	24
	rand_pad_8	0.1057	[0.0903, 0.1220]	62.4%	79.6%	96
Llama-3.2-3B	baseline	0.2488	[0.2106, 0.2922]	–	0.0%	96
	bucket_8	0.0336	[0.0219, 0.0467]	86.5%	88.6%	96
	pad_32	0.1560	[0.1290, 0.1849]	37.3%	565.1%	96
	batch_2	0.0760	[0.0640, 0.0880]	69.5%	0.0%	48
	batch_4	0.0817	[0.0610, 0.1029]	67.2%	0.0%	24
	rand_pad_8	0.1133	[0.0938, 0.1336]	54.5%	85.5%	96

No sample in any defended model–configuration pair had $\phi > 0.5$. For zero such cases among 30 samples, the Wilson 95% upper bound is 11.4% under a binomial sampling interpretation. This bounds a descriptive lexical-threshold event under that assumption, not a disclosure probability. The pilot establishes lower recorded scores for the evaluated frozen pipeline; it does not determine the remaining information or the performance of an adapted attacker.

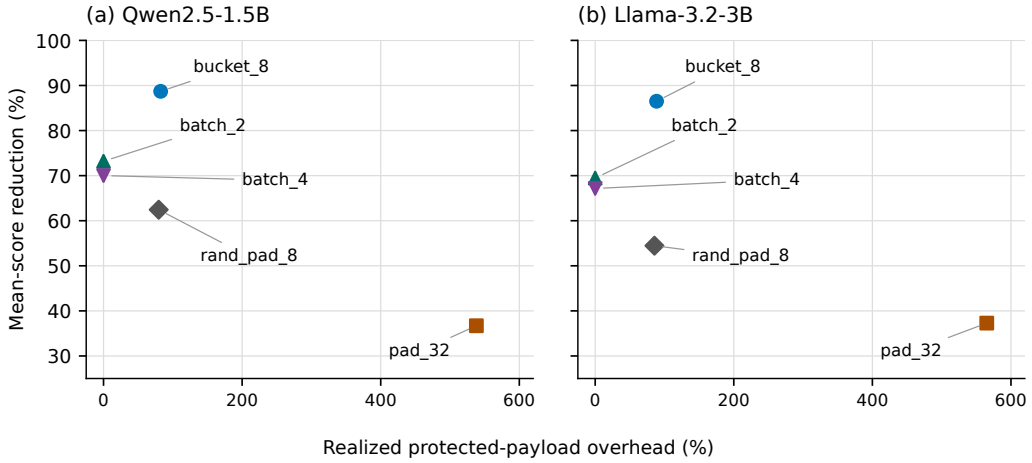


Figure 4. Mean lexical-score reduction versus realized protected-payload overhead, shown separately for the two 96-token pilot models ($n = 30$ prompts each; up to three segments, three samples per segment, 80-token T5 input and output limits). Identical axes show every model–transformation result from Table 11. Both coordinates use the saved completed runs, including the realized randomized padding. Batching has zero payload-sum overhead and changes the observable-unit count; transport overhead and latency were not measured. These are frozen-pipeline point estimates, not information-theoretic privacy estimates.

4.7. Decoder-Seed Stability and Adaptation with a MAP Inverse

To assess decoder-seed sensitivity, each condition was repeated with five seeds. Table 12 reports all 22 model–defense–attacker settings. Each setting contains 30 prompts evaluated with five seeds, giving 150 reconstructions. The same seed was used across conditions for a given prompt. The 512-token victim budget defines the available trace, but only the first heuristic segment was submitted for reconstruction; its output was scored against the full stored response, not a first-sentence reference.

For the unmodified baseline traces, mean ϕ for the first-segment output against the full reference was 0.1988 for Qwen2.5-1.5B and 0.1860 for Llama-3.2-3B. Across all settings, the standard deviation of the five seed-specific means ranged from 0.0032 to 0.0184. Within these two-model, 30-prompt comparisons, variation over the five seeds was smaller than the observed frozen-pipeline score changes. This finding does not address victim-generation variability or validate equal input coverage across transformations.

When no defense-specific adaptation was applied, `bucket_8` produced the largest reductions in mean ϕ: 78.8% for Qwen and 69.4% for Llama. Batching reduced the score by 36.8–49.5%, while randomized padding reduced it by 37.5% for Qwen and 43.8% for Llama. Fixed-width `pad_32` instead had a higher mean score than baseline, corresponding to signed reductions of -19.2% and -22.1%. A negative reduction denotes increased lexical overlap. The frozen `pad_32` encoder input is identical across both models and all 300 reconstructions; all 150 cross-model pairs with common prompt and seed indices produce identical predictions. Their different mean scores arise from scoring those predictions against different references. Response-specific trace-length variation does not reach this decoder.

The pattern changed after the prompt-disjoint MAP inverse was added. It was fitted on 270 separate calibration prompts per model and used to estimate original lengths before the unchanged T5 decoder was applied. Calibration and test prompts are exact-text disjoint but share scenario templates, so this evaluates new prompt variants within the represented scenarios. For `bucket_8`, the adaptive mean scores reached 0.2167 for Qwen and 0.2075 for Llama, meeting or exceeding their frozen baselines. Adaptive `batch_2` also exceeded

baseline for both models. With `batch_4`, the adaptive score was close to the Qwen baseline (a signed reduction of 2.8%) and above the Llama baseline (a signed reduction of −53.4%).

Adaptation had a different effect on the remaining transformations. For `pad_32`, the MAP inverse also yields a constant retained input within each model. Applying it reduced mean ϕ to 0.1583 for Qwen and 0.1492 for Llama, which corresponds to reductions of 20.4% and 19.7% relative to baseline. For `rand_pad_8`, part of the score lost by the frozen reconstructor was recovered, although the adaptive scores remained below baseline by 8.8% and 19.3%, respectively.

All 3300 MAP predictions exactly match the retained earlier predictions when aligned by model, prompt, condition, attacker and decoder seed. The current table uses newly recomputed crossed-bootstrap intervals with explicitly bound bootstrap seeds. These findings describe a sensitivity analysis of the evaluated pipeline. A defense-aware inverse recovered much of the lexical score lost under bucketing and batching, but inversion also changes the sequence subsequently segmented and truncated by T5. With 80-token limits on both input and output, equal victim-token budgets do not imply equal encoder coverage or comparable sentence boundaries. The adaptive procedure used an empirical length inverse followed by the unchanged T5 decoder, without end-to-end retraining on defended traces. Its scores therefore characterize this pipeline under limited adaptation; they do not isolate a defense's information loss or establish the performance of a fully retrained attacker.

Table 12. Five-seed evaluation on logged traces from a 512-token victim budget ($n = 30$ prompts per model; 150 reconstructions per setting). Only the first heuristic segment is decoded, with 80-token T5 input and output limits, and scored against the full response reference. Segmentation and encoder coverage differ across transformations. MAP calibration uses 270 exact-prompt-disjoint traces per model but shares scenario templates. Intervals are crossed bootstrap 95% CIs over prompts and seeds; SD describes the five seed-specific means. Reduction is $100(1 - \bar{\phi}/\bar{\phi}_{\text{baseline}})$; negative values denote scores above baseline.

Model	Defense	Attacker	Mean ϕ	95% CI	Seed SD	Reduction (%)
Qwen2.5-1.5B	baseline	frozen	0.1988	[0.1697, 0.2304]	0.0090	0.0
	bucket_8	frozen	0.0422	[0.0350, 0.0502]	0.0040	78.8
	bucket_8	adaptive	0.2167	[0.1927, 0.2413]	0.0042	−9.0
	pad_32	frozen	0.2371	[0.2108, 0.2638]	0.0088	−19.2
	pad_32	adaptive	0.1583	[0.1349, 0.1824]	0.0093	20.4
	batch_2	frozen	0.1200	[0.1056, 0.1354]	0.0070	39.7
	batch_2	adaptive	0.2380	[0.2114, 0.2666]	0.0184	−19.7
	batch_4	frozen	0.1004	[0.0804, 0.1248]	0.0165	49.5
	batch_4	adaptive	0.1932	[0.1635, 0.2225]	0.0123	2.8
	rand_pad_8	frozen	0.1242	[0.1034, 0.1449]	0.0035	37.5
	rand_pad_8	adaptive	0.1813	[0.1554, 0.2074]	0.0067	8.8
Llama-3.2-3B	baseline	frozen	0.1860	[0.1594, 0.2120]	0.0150	0.0
	bucket_8	frozen	0.0569	[0.0474, 0.0665]	0.0032	69.4
	bucket_8	adaptive	0.2075	[0.1816, 0.2341]	0.0165	−11.6
	pad_32	frozen	0.2270	[0.2021, 0.2528]	0.0067	−22.1
	pad_32	adaptive	0.1492	[0.1278, 0.1731]	0.0035	19.7
	batch_2	frozen	0.0999	[0.0859, 0.1157]	0.0091	46.3
	batch_2	adaptive	0.2139	[0.1826, 0.2452]	0.0046	−15.0
	batch_4	frozen	0.1175	[0.1001, 0.1360]	0.0052	36.8
	batch_4	adaptive	0.2852	[0.2512, 0.3214]	0.0145	−53.4
	rand_pad_8	frozen	0.1044	[0.0864, 0.1227]	0.0072	43.8
	rand_pad_8	adaptive	0.1500	[0.1230, 0.1784]	0.0075	19.3

5. Discussion

5.1. Evidence of Trace-Dependent Recovery

The strongest control evidence concerns replacing the trace by a constant: both models lose lexical score, and shuffling also reduces Qwen's score after correction. Neither cross-prompt contrast is conclusive, and Llama's shuffled contrast is inconclusive after Holm correction. Thus the experiments support sensitivity to selected trace features, not a universal advantage of the target trace over every partial ablation. Controls preserve response-dependent quantities such as length, mean or histogram and can change segmentation. Their residual scores cannot be assigned entirely to the decoder's language prior. Estimating information-theoretic leakage would require a defined response variable, repeated sampling and a suitable estimator.

5.2. Limits of the Cross-Model Comparison

Exact logged byte lengths remove the line-collector ambiguity from the reported reconstruction inputs. Reference quality remains a distinct issue: archived TinyLlama and Phi-3.5-mini fragments have malformed word boundaries, whereas the new two-model cohorts pass explicit full-sequence decoding checks. For fixed newly recorded token IDs, legacy decoding reproduces the archived inputs and references exactly, while full-sequence decoding restores word-boundary spaces. With reconstruction settings and per-sample seeds held constant, both resulting score increases have paired scenario-bootstrap intervals excluding zero. This effect jointly includes the changed length trace and changed scoring reference; neither contribution is isolated. The archived low scores therefore cannot support a general conclusion of poor reconstructor transfer to these two models.

The five other archived-reference series support conditional comparisons of the fixed reconstructor on their emitted texts, not an intrinsic model-vulnerability ranking. Historical token IDs and exact victim revisions are unavailable, while tokenizer vocabulary, response style, refusal behavior and compatibility with T5 training data can differ. The matched two-model rendering experiment is consequently reported separately from those five series. The synthetic corpus also contains only 60 scenarios, and the MAP test and calibration sets share those scenarios despite exact-prompt disjointness.

5.3. Practical Exploitability

For the largest mean in the archived-reference cohort, LR@0.5 is 65.33% and ROUGE-1 precision is 0.2921. Neither quantity is a disclosure rate: lexical overlap does not verify an entity, proposition or actionable answer. No semantic annotation or task-specific disclosure evaluation was performed. Practical harm might involve recovered phrases, entities, refusals or topic cues, but the present metrics do not quantify those outcomes. Deployment-facing conclusions additionally require extraction of observables from encrypted traffic and validation under transport buffering and multiplexing.

Reconstruction performance, transport observability, and mitigation performance require separate evidence. The MAP inverse recovered score losses on bucketing and batching within the evaluated pipeline, but it also changes segmentation and the amount of input retained under the 80-token encoder limit. The study therefore measures the combined effect of the length transform, inverse, preprocessing, and fixed decoder. For `pad_32`, the complete observable trace retains its number of positions, but the actual retained encoder input is constant within each condition. Its positive lexical scores are consequently not evidence that this decoder exploits response-specific length variation. This finding concerns $E(Z)$ and does not establish absence of information in Z for other attackers. A comparison that fixes observation coverage across conditions would help isolate these

mechanisms. End-to-end training on defended traces and disclosure-oriented human or task evaluations remain separate extensions.

5.4. Limitations

The three-candidate experiments retain a selector that includes PAD scores after EOS. The bounded replay establishes actual selection changes under EOS masking, but does not estimate their frequency across the campaign or their effect on full-response scores. Later segments condition on the selected text, so corrected full-response results require a separate end-to-end campaign with the revised selector and retained candidates. Current score rankings, confidence intervals, rendering contrasts and defense comparisons are conditional on the original implementation. The one-candidate MAP results are unaffected by this selection mechanism.

The observation is an instrumented log oracle. No packet capture or experiment with TLS, HTTP/2 or QUIC demonstrates that its token boundaries are available to a passive observer. The two historical reference-format diagnostics remain in the archived cohort for traceability, and the other archived references lack original token-ID provenance. Newly generated references are verified for TinyLlama and Phi-3.5-mini, but their configurations are not certified replicas of the historical victim executions.

The primary cohorts use one assembled reconstruction per prompt with three stochastic samples per segment. Decoder-seed uncertainty is evaluated separately for two models and 30 prompts; five seeds are not five independent prompt samples. Scenario bootstrap accounts for grouping within the 60 represented scenarios, without establishing generalization beyond the synthetic corpus. MAP calibration and test inputs are disjoint as exact prompt strings, not as scenarios. Pilot decoder seeds are condition-specific, and their single realizations cannot separate all stochastic variation from transformation effects.

The main archived responses reach the token limit in 34.3–100% of cases; both historical and new Phi-3.5-mini cohorts lack EOS-complete responses. Reconstruction processes all heuristic segments in the primary and new-reference cohorts, up to three in the 96-token pilots, and one in MAP. All use the full stored response as the scoring reference. Transform-dependent segmentation and the 80-token encoder limit prevent treating these scopes as matched information budgets. The MAP inverse is not end-to-end retraining, and payload overhead excludes transport syntax, buffering delay and latency. Finally, the historical Gemma cohort uses a different victim-generation implementation from the other six archived cohorts.

5.5. Ethical Considerations

The study used synthetic prompts and locally generated responses; it did not intercept user traffic or process identifiable personal data. The artifact release is intended to support defensive evaluation of streaming implementations. Deployment testing should use authorized systems, controlled accounts, and coordinated disclosure when a service-specific weakness is found.

6. Conclusions

The completed experiments comprise 6600 T5 reconstructions and 600 newly generated victim responses. Reconstruction from validated logged lengths yields mean lexical scores of 0.482–0.535 for five archived-reference cohorts. For TinyLlama and Phi-3.5-mini, verified full-sequence decoding yields means of 0.435 and 0.504. Legacy rendering of the newly saved token IDs exactly reproduces the corresponding archived references and length vectors. Under matched T5 settings and seeds, changing the rendering of both the input trace and scoring reference increases mean ϕ by 0.3560 and 0.3831. This computational

comparison identifies observation and reference construction as consequential parts of the evaluation; it does not separate their individual contributions or establish an intrinsic model-vulnerability ranking.

The partial-ablation controls support sensitivity to selected trace features: three of six contrasts remain significant after Holm correction, while neither cross-prompt contrast is conclusive. In the 96-token pilots, all ten model–transformation contrasts show significant score reductions after correction, ranging from 36.7% to 88.7%. These reductions describe the tested frozen reconstructor and realized transformations. The first-segment, five-seed study shows that an exact-prompt-disjoint MAP length inverse can recover much of the loss under bucketing and batching, although calibration shares scenario templates with the test set.

These conclusions are conditional on the recorded implementation: the three-candidate experiments use the original PAD-inclusive selector, whereas MAP uses one candidate. Constant `pad_32` encoder inputs also show why positive lexical overlap need not demonstrate use of response-specific trace variation. Practical privacy claims require evidence beyond these lexical scores. Transform-dependent segmentation and the 80-token encoder limit change the retained input coverage, and payload-size accounting excludes transport costs and latency. Evaluations with matched observation coverage, unseen scenarios, extraction from encrypted traffic, and disclosure-oriented outcomes are needed to establish the deployment significance of these results.

Appendix A. Artifact Provenance and Completed Experiments

Supplementary Data S1 is the frozen local package `LLeakM-S1-20261101-v1`, supplied as a compressed archive with a SHA-256 manifest, a README and a CPU-only verification entry point. It contains the completed records, analysis code, source logs, model revision identifiers and manuscript files; model weights are obtained separately from the pinned checkpoints. The archive is prepared for accompanying submission and can be requested from the corresponding authors; public archival of this version remains pending. The source project repository is https://github.com/jackaugustyn/LLeakM. Table A1 identifies the original 512-token source directories. Results reside under `experiment_validation/revalidation_20261031/`: `inference/cuda_full/` contains the 2100 reconstructions from canonical archived lengths, `secondary/cuda_full/` the 3900 control, defense and MAP records, and `inference/fresh_tinyllama_cuda/` and `inference/fresh_phi_cuda/` the two 300-response new-reference reconstructions. Their generation artifacts are under `fresh_references/`.

Each completed run binds its sample inventory, source hashes, model files, code, seeds and environment to a protocol. Independent checks verify complete unique keys, saved text metrics, encoder input tokenization, segment coverage and cohort counts. All planned reconstructions are present; partial probes are excluded. The `inference_validation/` reports and the secondary summary contain the numerical sources for the manuscript, while `final_review_20261101/` records the independent numerical audit. The `article_revision_latest_20261101/` directory records the updated table and figure checks, plotting inputs, reproducible figure generator and manuscript build.

The `review_revision_20261101/` directory records the present source-bound edits, the candidate-selection audit, exact local-MAP and seed checks, and the new manuscript validation. The supplementary manifest binds the files actually distributed in S1; its verification entry point checks those hashes, inventory completeness and per-record lexical cosine without loading a model. Version-pinned reconstructor model cards and their retrieved text are included in the source-check evidence.

The earlier `analysis/repro_manifest.json` and `analysis/revision_snapshot_20261031.json` remain historical inventories and are not attestations of the new results. `completed_manifest.json` records the completed experiment artifacts before their integration into the manuscript; it is an experiment inventory, not a hash inventory of the integrated manuscript. These provenance records cannot retroactively establish historical victim revisions or undocumented old reconstruction inputs.

Table A1. Source-response directories for the archived-reference cohort ($n = 300$ per model). New reconstructions use validated log lengths; TinyLlama and Phi-3.5-mini remain reference-format diagnostics in this cohort.

Model	Artifact directory
Qwen2.5-1.5B-Instruct	`run_full_20260920_qwen_1_5b_full`
Qwen2.5-3B-Instruct	`run_full_20260920_qwen_3b_full`
Phi-1.5	`run_full_20260920_phi_1_5_full`
Llama-3.2-3B-Instruct	`run_full_20260920_llama_3_2_3b_full`
Gemma-2-2B-it	`run_full_20260920_gemma_2_2b_full`
TinyLlama-1.1B-Chat-v1.0	`run_full_20260920_tinyllama_1_1b_full`
Phi-3.5-mini-instruct	`run_full_20260920_phi_3_5_mini_full`

Appendix B. Local Length Inversion and Random Seeds

Appendix B.1. Empirical Length Inverse

Let $c(x)$ count individual clean lengths in the calibration corpus, and let $S = \{x : c(x) > 0\}$. For an unbatched defense, the calibration pass counts $C_z(x)$, the occurrences of clean length x aligned with defended value z. Deterministic transformations use one pass; `rand_pad_8` uses 16 independent padding realizations per calibration trace. The inverse returns the most frequent x in C_z, resolving ties in favor of the smaller length. An unseen z returns the global mode of c, with the same tie rule. Thus the estimator is local to an observed unit. The baseline bypasses inversion.

For batching with maximum size $k \in \{2, 4\}$, calibration counts tuples in non-overlapping consecutive groups, including each final partial group. Let $C_{z,s}(\mathbf{x})$ count a tuple of size s and sum z, and let $T(s)$ count calibration traces whose final group has size s (including full final groups). Interior observed groups are assigned size k. For the last observed group, the implemented heuristic chooses

$$\widehat{s} = \arg\max_{1 \le s \le k} \left\{ \left[\sum_{\mathbf{x}} C_{z,s}(\mathbf{x}) \right] \max(1, T(s)) \right\},$$

with ties resolved in favor of the larger s. This product of counts is a size-selection heuristic, not a separately validated posterior for the final group.

For the chosen (z, s), the inverse returns the most frequent recorded tuple, with lexicographically smallest tuples winning ties. If that pair was never observed, dynamic programming finds a tuple in S^s with sum z maximizing

$$\sum_{j=1}^{s} \log \widetilde{p}(x_j), \qquad \widetilde{p}(x) = \frac{c(x) + 1}{\sum_{u \in S} c(u) + |S|}.$$

The recurrence appends one supported length at a time, retains the best tuple for each attainable subtotal, and rejects subtotals exceeding z; equal floating-point scores use the lexicographic tie rule. If the requested sum is unattainable, it repeats the global mode s

times, which need not preserve z. These rules specify the implemented local inverse and do not constitute globally optimal sequence inference.

Appendix B.2. Decoder, Transformation and Bootstrap Seeds

Define $H(a_1, \ldots, a_m)$ by converting each argument with Python's `str`, joining the strings with byte `0x1f`, UTF-8 encoding, and applying BLAKE2s with a four-byte digest. Interpret the digest as an unsigned big-endian integer and apply bitwise AND with `0x7fffffff`. This is the artifact function `stable_seed`; it is independent of Python's randomized `hash()`.

The immutable main and verified-reference inventories store each exact seed in the `reconstruction_seed` field. The matched renderings use the same recorded seed for each prompt. For the two pilots, the decoder seed is

$$H(\texttt{secondary-pilot-decoder-v1}, 20260706, \texttt{label}, i, \texttt{suite}, \texttt{condition}),$$

where i is the recorded prompt index and the suite is `trace_control` or `defense_pilot`. Thus the two pilot baselines do not share decoder seeds. The defense transformation seed is $H(\texttt{secondary-pilot-trace-v1}, 20260706, \texttt{label}, i, \texttt{condition})$; shuffling instead uses $20260706 + 17i$.

For MAP, decoder seed $H(\texttt{decoder}, r, i)$ is reused across models, defenses and attacker modes for a given prompt index and seed index $r \in \{20260706, \ldots, 20260710\}$. Transformation seed $H(\texttt{trace}, 20260920, \texttt{label}, i, \texttt{condition})$ fixes a trace realization across decoder repeats and attacker modes. The inverse is fitted with seed $a = H(20260920, \texttt{label}, \texttt{condition})$; calibration trace index j and padding draw d use $H(\texttt{calibration}, a, \texttt{condition}, j, d)$, with zero-based indices.

The main and verified-reference aggregators initialize their NumPy generator with `20260706`, consuming it in the pinned aggregation order. For every pilot or MAP cell, the bootstrap generator instead uses $H(\texttt{bootstrap}, \texttt{suite}, \texttt{label}, \texttt{condition}, \texttt{attacker})$; its paired contrast with baseline uses the same arguments with `contrast` replacing `bootstrap`. Each generator is initialized separately. Pilot intervals resample 30 prompt scores; MAP intervals independently resample the 30 prompt indices and five seed indices, then average their Cartesian-product submatrix. All use 10,000 replicates and the 2.5th and 97.5th percentiles. The matched-rendering contrast separately uses seed `20261101` and 60 paired scenario means. Exact key strings, inventories and aggregation order are retained in the supplementary code and protocols.

Appendix C. Bounded Candidate-Selection Audit

The completed three-candidate campaign did not retain every sampled output ID and transition score. We therefore replayed 24 first-segment inputs with the original first-segment checkpoint, Transformers 4.46.3, float32 CUDA execution and saved per-record seeds. Before inference, selection was fixed to input-length extrema for two inputs from each of the seven archived and two verified-reference cohorts, plus six defense-pilot inputs covering baseline, bucketing, fixed padding, both batching sizes and random padding. This deliberately selected diagnostic set is not a random sample of the campaign.

All 24 original selected texts were reproduced exactly. The audit retained all 72 candidate sequences and step scores, and compared the executed full-step sum with a sum through the first generated EOS, including that EOS and excluding subsequent PAD. For a sequence without EOS, all generated steps are retained. The two rules selected different text in 18 of 24 cases: 11 of 14 archived inputs, three of four verified-reference inputs and four of six defense-pilot inputs. Post-EOS tails occurred in 38 of 72 candidates; each such

tail included a nonfinite score, while no nonfinite score occurred before or at EOS. The normalization uses the sampling-processed scores, not the unconstrained model logits.

These findings establish that the defect can affect actual generated candidates. They do not estimate its prevalence over the campaign, a corrected full-response mean or the direction of change in any table. The original selector remains the definition of the three-candidate results reported here. Revising that definition requires replay of later segments because their context depends on the selected text. The one-candidate MAP study does not use this ranking. The supplementary `review_revision_20261101/ranking_audit/` directory contains the fixed input selection, complete candidate records, EOS-masking tests, protocol and audit report.

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Data Availability Statement: The source project repository is https://github.com/jackaugustyn/LLeakM. The frozen local Supplementary Data S1 package, `LLeakM-S1-20261101-v1`, contains completed reconstructions, generated token IDs, source logs, reference-validation records, analysis code and a file-level SHA-256 manifest with a verification entry point. It is prepared for accompanying submission and available from the corresponding authors on request; public archival of this completed version remains pending. Appendix A identifies its version, contents and provenance boundaries.

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